

Topics: 16.7 Surface Integrals; 16.9 The Divergence Theorem

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1. Evaluate the following scalar surface integrals.

(a) $\iint_S x^2 y z \, dS$, where S is the part of the plane $z = 1 + 2x + 3y$ that lies above $[0, 3] \times [0, 2]$.

(b) $\iint_S y^2 z^2 \, dS$, where S is the part of the cone $y = \sqrt{x^2 + z^2}$ with $0 \leq y \leq 5$.

(c) $\iint_S z \, dS$, where S is given by $\mathbf{r}(u, v) = u^2 \mathbf{i} + v \mathbf{j} + u \mathbf{k}$ as (u, v) ranges over $[0, 1] \times [0, 1]$.

2. Evaluate the flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for the given vector field $\mathbf{F}$ and oriented surface S .

(a) Let $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$ and let S be the surface parametrized by $\mathbf{r}(u, v) = \langle u, v, u^2 + v^2 \rangle$, for $0 \leq u \leq 1$ and $0 \leq v \leq 1$, with upward orientation.

(b) Let $\mathbf{F}(x, y, z) = \langle y, -x, 2z \rangle$ and let S be the spherical cap $x^2 + y^2 + z^2 = 4$ with $z \geq 1$, with downward orientation.

3. Use the Divergence Theorem to find the outward flux of $\mathbf{F}$ across the closed surface S .

(a) Let $\mathbf{F}(x, y, z) = \langle xye^z, xy^2z^3, -ye^z \rangle$ and let S be the boundary surface of the rectangular box bounded by the coordinate planes and the planes $x = 3$, $y = 2$, and $z = 1$.

(b) Let $\mathbf{F}(x, y, z) = \langle xe^y, z - e^y, -xy \rangle$. Let S be the ellipsoid $x^2 + 2y^2 + 3z^2 = 4$.

(c) Let $\mathbf{F}(x, y, z) = z\mathbf{i} + y\mathbf{j} + x\mathbf{k}$ and S be the boundary surface of the upper half-ball $x^2 + y^2 + z^2 \leq 1$, with $z \geq 0$, including the disk in the plane $z = 0$.

(d) Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + y^2\mathbf{j} + z\mathbf{k}$ and let S be the boundary surface of the tetrahedron bounded by the plane $x + 2y + z = 2$, and the three coordinate planes.

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Surface Integrals of Scalar Functions. Suppose f is a function of three variables whose domain includes a smooth surface S . If S is given parametrically by

$$\mathbf{r}(u, v) = x(u, v) \mathbf{i} + y(u, v) \mathbf{j} + z(u, v) \mathbf{k},$$

and S is covered only once as (u, v) ranges over D , then

$$\iint_S f(x, y, z) \, dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| \, dA.$$

If S is given explicitly by $z = g(x, y)$ for $(x, y) \in D$, then

$$\iint_S f(x, y, z) \, dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} \, dA.$$

Surface Integrals of Vector Fields. Let S be an oriented surface with unit normal vector $\mathbf{n}$. The flux of a vector field $\mathbf{F}$ across S is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n} \, dS.$$

If S is parametrized by

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}, \quad (u, v) \in D,$$

and is covered only once, then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, dA,$$

using the orientation determined by $\mathbf{r}_u \times \mathbf{r}_v$. For the opposite orientation, use $-(\mathbf{r}_u \times \mathbf{r}_v)$. If S is the graph $z = g(x, y)$ with $(x, y) \in D$, then for upward orientation,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(x, y, g(x, y)) \cdot \langle -g_x, -g_y, 1 \rangle \, dA.$$

For downward orientation, instead use $\langle g_x, g_y, -1 \rangle$.

The Divergence Theorem. Let E be a simple solid region, or a finite union of simple solid regions, and let S be the boundary surface of E , given with the positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous first-order partial derivatives on an open region containing E . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} \, dV.$$

Suggested Textbook Problems

Section 16.7: 1-32

Section 16.9: 1-17, 19-24